PCA Exit Ticket

Ronny Kiprono - CMAT 651

Time: 5 minutes Work alone. No calculator.

1. Concept

PCA gives the following eigenvalues:

$\lambda_1 = 3, \quad \lambda_2 = 1$

a) Find the total variance.

b) Find the percentage of variance explained by PC_1.

c) Which principal component should be retained if only one component is allowed?

2. Python by Hand

Complete the missing line from the PCA code used in class:

```
pca = PCA(n_components=2)
PC = _____
```

What does n_components=2 tell Python to keep?

3. Interpretation - Student Dropout Dataset

Suppose PCA on the student-performance variables gives:

$PC_1 = 70\%, \quad PC_2 = 20\%$

a) How much variance is explained by PC_1 and PC_2 together?

b) The original dataset uses four numerical variables:

X_1, X_2, X_3, X_4

After PCA, we retain PC_1 and PC_2. Write the dimension reduction:

_____ → _____

c) In our code, why is Target added after PCA rather than included in the PCA input?

PCA Exit Ticket - Answer Key

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1. Concept

a) Total variance:

$\lambda_1 + \lambda_2 = 3 + 1 = 4$

b) Variance explained by PC_1:

$\lambda_1 / (\lambda_1 + \lambda_2) = 3/4 = 75\%$

c) Retain PC_1 because it has the largest eigenvalue and captures the most variance.

2. Python by Hand

```
PC = pca.fit_transform(X_scaled)
```

n_components=2 means PCA keeps two principal components: PC_1 and PC_2.

3. Interpretation

a) 70% + 20% = 90%

b) Dimension reduction:

$4 \rightarrow 2$

c) Target is the student outcome. PCA is performed on the numerical input variables without using the outcome. Target is added afterward so the PCA scores can be compared across Dropout, Enrolled, and Graduate groups.